3.0 Introduction

The road conditions are not uniform whenever we are in travel in the vehicle. Moreover we do not drive the vehicle always at the same speed. The vehicle must be operated at lower speed when it needs more pulling power. Moreover based on the operating conditions, the vehicle requires different speed and torque. To perform this, a box containing different gears, shafts and bearings is required. This box is called as the gear box. It can be also called as teeth wheel box. By using this box, the driver can achieve different speeds and torques for a vehicle. Gear box is the one which comprises of various teethed gears for different vehicle speed conditions.

3.1 Gear box location

Gear box is located at the second place of the vehicle's transmission system. In the vehicle after the engine, clutch is located. After the clutch gear box is located. It is located in between the clutch and the propeller shaft.

3.2 Gears

The commonly used gears and their arrangements can be seen in this section. With a help of teethed wheels we can transmit the energy in rotational, parallel and in perpendicular directions. In machines gears are used in places wherever the transmission of rotational force is taking place. Moreover the gears are used to transmit the rotational force without any slip. It is used in places where there is a small distance for transmitting the rotational force (torque).

3.2.1 Types of gears

Gears are of many different types. The following are some of the important gears

1. Spur gear
2. Helical gear
3. Double helical gear
4. Bevel gear
5. Rack and pinion gear
6. Worm and worm gear
7. Internal gear
8. Sprocket wheel

3.2.1.1 Spur gear

It is like a circular shaped one made on a cylindrical metal having teeth cut in parallel to the axis of the cylinder. It is called as spur gear.



Figure 3.2.1.1 Spur gear

This gear is used to transmit the rotary power only between two parallel shafts. It is capable of transmitting moderate force and speed. A smaller gear is called as pinion. Normally in gear box this type of gear is used for first and reverse gear speed operations.

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